

# PRAVEEN K

SOC Analyst | Junior Threat Intelligence Analyst | Cybersecurity Analyst

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## PROFESSIONAL SUMMARY

Entry-level SOC Analyst with proven hands-on depth across the full detection-to-response lifecycle - built a home lab from scratch simulating real-world MITRE ATT&CK kill chain scenarios including brute force, lateral movement, and C2 beaconing. Engineered Pherion, a Python-based IDS with 37+ MITRE-mapped detection rules and Splunk integration, achieving ~30% false positive reduction through ML-based anomaly detection. Skilled in IOC/IOA analysis, PCAP investigation, Sigma rule authoring, alert triage, and SIEM-based incident correlation. Certified across Cisco, Fortinet, and Palo Alto Security Operations. Targeting L1 SOC Analyst roles at MSSPs and GCCs to contribute to 24/7 security operations and incident response.

## TECHNICAL SKILLS

**SIEM & Monitoring** : Splunk (SPL, log ingestion, alert rules, dashboards), Microsoft Sentinel , QRadar , SIEM Monitoring, Alert Triage, Log Analysis, Incident Ticketing.

**Security Tools** : Wireshark, Nmap, Burp Suite, Nessus, Metasploit, Kali Linux

**Threat Detection & Response** : IDS/IPS, IOC/IOA Analysis, Threat Hunting, Threat Intelligence,Malware Analysis, Incident Response, PCAP Analysis, Vulnerability Assessment, RCA (Root Cause Analysis), Incident Escalation, Playbook/Runbook Execution, SOAR,Threat Feeds,OSINT.

**Frameworks & Standards** : MITRE ATT&CK, Cyber Kill Chain, OWASP Top 10, NIST Cybersecurity Framework , IAM Fundamentals, CVSS Scoring.

**Networking & Protocols** : TCP/IP, DNS, HTTP/HTTPS, VPN, Firewalls, ARP, DHCP, Packet Analysis, Network Scanning, Port Enumeration.

**OS & Platforms** : Kali Linux, Windows Security, Linux CLI, VirtualBox,Virtual Lab Environments.

**Programming & Scripting** : Python (security scripting, automation), Bash/Shell Scripting,PowerShell (basics).

**Concepts** : AES-256 Encryption, Steganography, Zero Trust Architecture, EDR (CrowdStrike/SentinelOne),ServiceNow/Jira ticketing, 24x7 SOC operations

## PROJECTS

### Pherion – AI-Powered IDS & SOC Monitoring Platform

**Tech Stack:** Python, Scapy, Scikit-learn, Pandas, SQLite, JSON, Splunk

- Engineered 37+ detection rules mapped to MITRE ATT&CK TTPs (T1046 – Port Scan, T1071 – C2 Beaconing, T1110 – Brute Force, T1557 – ARP Spoofing, T1048 - DNS Tunneling, T1021 - Lateral Movement) for structured threat intelligence reporting.
- Integrated ML-based anomaly detection with auto-tuning thresholds, reducing false positive alert noise by ~30% in lab testing - improving analyst triage efficiency.
- Generated SIEM-ready JSON logs ingested into Splunk, enabling real-time alert investigation and SOC workflow simulation with dashboard visibility.

### Praxion – USB Malware Scanner & Threat Detection Tool

**Tech Stack:** Python, YARA, PE File Analysis, VirusTotal API, ssdeep

- Automated IOC identification using VirusTotal API and ssdeep fuzzy hashing, enabling faster malware triage and threat classification decisions.
- Built YARA rule matching engine with heuristic scanning and PE inspection for detecting known and unknown malware families on USB devices.
- Implemented forensic-grade logging and secure quarantine system, producing structured evidence output suitable for incident documentation.

### PRISM – SIEM Rule Generator from Raw Logs

**Tech Stack:** Python, Rich, Regex, Sigma YAML, Splunk SPL, JSON, MITRE ATT&CK

- Architected a regex parsing pipeline for auth.log, Apache access logs, and Windows Security events - detecting failed logins, port scan surges, and privileged execution attempts per source IP against defined thresholds.
- Engineered automated rule translation layer converting raw log insights into deployable Sigma YAML and Splunk SPL detection rules, with a built-in validation module cross-referencing generated rules against sample malicious logs before deployment.
- Designed a MITRE ATT&CK mapping engine assigning tactics and CVE identifiers to extracted artifacts, with normalized JSON output for universal SIEM ingestion.

## HOME LAB – SOC ANALYST SIMULATION ENVIRONMENT

**Tech Stack:** VirtualBox, Kali Linux, Windows 10, Splunk Enterprise, Wireshark

- Built a segmented dual-VM SOC lab (Kali Linux / Windows 10 + Splunk Enterprise) simulating real-world attack chains across T1110 (Brute Force), T1078 (Valid Accounts), T1046 (Port Scan), and T1021 (Lateral Movement) - generating 50+ log events for triage practice.
- Executed full L1 analyst workflow: alert detection → Windows event log investigation → IOC correlation → findings escalation → structured incident documentation in simulated ServiceNow/Jira ticketing format across 10+ incident scenarios.
- Authored custom Splunk SPL detection rules with threshold-based false positive tuning; executed NIST IR playbook (Identify → Contain → Eradicate → Recover → Document) for each simulated incident, producing written RCA reports.

## THREAT INTELLIGENCE & CVE RESEARCH

### CVE-2021-44228 | Log4Shell RCE Analysis

- Analyzed Log4Shell RCE flaw in Apache Log4j (v2.0–2.14.1) where malicious JNDI strings in HTTP headers trigger remote code execution; mapped attack chain: Initial Access (T1190) → Code Execution (T1059) → C2 Callback (T1071).
- Built Splunk query to detect JNDI patterns in web logs and alert on unexpected outbound LDAP traffic - simulating SOC L1 triage workflow.
- Documented patch (Log4j 2.15.0), WAF rules, and input sanitization as mitigations in a structured SOC analyst incident brief.

### Scattered Spider (UNC3944) – Threat Actor Analysis

- Analyzed the 2023 MGM/Caesars campaign – mapped full attack chain: MFA fatigue (T1621) → credential theft (T1078) → Okta abuse → lateral movement (T1021) → ransomware (T1486) across MITRE ATT&CK, producing a SOC L1 threat intelligence brief.
- Built Splunk detection logic for MFA fatigue patterns and anomalous Okta logins; documented mitigations including adaptive MFA, UEBA alerting, and zero-trust access review aligned to NIST IR playbook.

## HANDS-ON TRAINING & INTERNSHIP

### Hackup Technology pvt.Ltd – Cybersecurity Internship (1 Month)

- Identified 12+ vulnerabilities (SQLi, XSS, IDOR) across 3 web applications via Burp Suite, mapping all findings to OWASP Top 10 with CVSS severity ratings and structured remediation documentation.
- Executed controlled exploitation chains on 4 vulnerable lab machines via Metasploit - achieving privilege escalation on 3 targets; analyzed post-exploitation traffic in Wireshark to identify IOC patterns and produced pentest reports with remediation steps.

### Vital Skills × Techkriti – IIT Kanpur - Ethical Hacking & Network Security Internship (1 Month)

- Executed end-to-end penetration tests across 5 lab targets following the full Cyber Kill Chain (Recon → Scan → Exploit → Post-Exploit); simulated MITM, brute-force, and sniffing attacks to map attacker TTPs against defensive controls.
- Conducted vulnerability assessments using Nmap and Burp Suite; analyzed 3 PCAP files in Wireshark to reconstruct attack paths and delivered risk-rated reports with prioritized remediation steps.

## EDUCATION

### B.E. Computer Science and Engineering

St.Michael College of Engineering and Technology - Kalayarkoil

2022 - 2026

## CERTIFICATIONS

- Cisco Networking Academy – Ethical Hacker
- Fortinet Training Institute – NSE 3 – Associate in Cybersecurity
- Cisco Networking Academy – Network Defence
- Palo Alto Networks – Security Operations Fundamentals

Dec 2025  
Oct 2025 - Oct 2027  
Jun 2026  
May 2026